FLEET TEAM DIGEST

737MAX-FTD-34-22008

Issue Title : Rockwell Collins Traffic Collision and Avoidance System (TCAS) TTR-2100 Resolution Advisory (RA) Display Nuisance Faults

Airplane Model	ATA	Minor Model(s)
737MAX	3445-00	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
737NG	9E991	

Originated Date	Last Revised Date	Created On
01/23/2023	11/05/2025	10/10/2025
Estimated Completion Date	**Next Update Date**	

Revision Description

Updated References, Applicability, Interim Action, Final Action, and Milestones to include information about the Collins Service Bulletin.

Applicability

737NG and 737 MAX airplanes utilizing Rockwell Collins TCAS Part Number PN: 822-2911-002 serial number 7B4LH, 7B4BJ and below with the exception of 7B4B6, 7B4B5, and 7B49M

Description

Boeing has received reports from multiple operators of TCAS FAIL messages appearing on the Captain's or First Officer's navigation display, with correlated maintenance message 34-45081 and/or 34-45091: TCAS RA 1/2 Status discrete input fault from Display Processing Computer (DPC). This message occurred during any phase of flight, but was mainly reported during taxi or shortly after takeoff, and could be cleared by performing a TCAS self-test after IRS alignment has completed, or by cycling the TCAS circuit breaker. This message is triggered to inform crews that the TCAS system has found a fault which prevents TCAS related information from being displayed to the flight crew. Analysis of TCAS computer units that were removed and returned to the manufacturer, Collins Aerospace, indicated the "Ready" signal coming from the DPCs were not being set causing the TCAS to indicate a system failure with no further faults logged in the unit's memory. This FTD provides detail and information on the root cause investigation being performed in coordination with the supplier of this unit, Rockwell Collins.

Status

Boeing has been working with operators to gather relevant information, such as crew or maintenance actions before and event. Additionally, an AHM Onboard report has been developed to record related airplane system data when an event occurs to aid in root cause determination.

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A review of the AHM Onboard Report data in coordination with Collins has found that this condition is usually caused by performing a TCAS self-test before the IRS system has completed alignment, as noted in Boeing Maintenance Tip (MT) 737 34-056 R3 and Collins SIL TTR-2100/4100 16-5.

Collection and analysis of data from the AHM Onboard report and testing performed by an operator, it was confirmed that performing a self-test after the IRS has completed alignment will clear the nuisance fault, even if a TCAS FAIL message is not currently being displayed.

Initial data from the AHM Onboard report indicated that the TCAS unit is declaring a latched fault mainly during power on or engine start, with the TCAS FAIL message appearing sometime later, depending on when the control switch is placed in the TA/RA mode of operation.

Additional data collection and analysis has found that the root cause of these events is due to a TCAS self-test being initiated while the IRUs are not aligned. Currently two scenarios have been found in the data. In the first scenario, the TCAS self-test is being activated on the prior flight just after landing and parking. As the crew is preparing to depart the airplane, the IRU control knobs are turned to the Align/Off position and the TCAS control knob is turned to the Standby position, at which time a TCAS self-test is being activated. The TCAS self-test activation may be accidental, however, this causes the fault to latch and a TCAS FAIL message will appear during the next flight, unless a TCAS power cycle is performed, or another TCAS self-test is activated after the IRUs have completed alignment. In the second scenario, a TCAS self-test is being activated in between flights, about 20 to 30 minutes after landing and parking. At this time, the IRUs have already been turned to the align/off position, while waiting for the next flight and the TCAS self-test button is pushed causing the fault to latch and the TCAS FAIL message to appear during the next flight, unless a TCAS power cycle is performed, or another TCAS self-test is activated after the IRUs have completed alignment.

Interim Action

Release of this FTD is to inform operators experiencing nuisance TCAS FAIL message with correlated MM 34-45081/91 to be be aware of the information provided in Boeing MT 737 34-056 R3 and Collins SIL TTR-2100/4100 16-5.

When a nuisance TCAS FAIL message with correlated MM 34-45081/91 occurs, the fault can be cleared by performing a TCAS self-test after the IRS system has completed alignment, as detailed in Boeing MT 737 34-056 R3. Please note a TCAS self-test

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can only be performed while the airplane is on the ground, as the self-test function is prohibited within the unit while airborne. Alternatively, operators have reported that cycling the TCAS circuit breaker while on the ground also clears these nuisance events. Replacement of the TCAS Computer unit is not necessary if the fault clears.

Boeing further recommends raising awareness to flight and maintenance crews to be careful while changing the TCAS control knob position so that the TCAS self-test button is not accidentally pressed.

Final Action

Boeing has confirmed root cause of these events were due to activation of a TCAS self-test while the IRUs were not aligned.

Operators are recommended to contact Collins to arrange installation of SB TTR-2100-34-14 to alleviate these events.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Initial AHM Onboard Report Deployed: SEP-2022 (Complete)

Initial Root Cause Established: OCT-2022 (Complete)

Updated AHM Onboard Report Deployed: APR-2022 (Complete)

Final Root Cause Established: 24-MAY-2023 (Complete)

Collins SB Available: 02-SEP-2025 (Complete)

Supplier Information

ROCKWELL COLLINS, INC.

DIV COLLINS AEROSPACE

400 COLLINS RD NE

CEDAR RAPIDS, IA 52498-0503

Reference Categories

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Reference Type	Reference Number	Revision Number	Reference Date
Service Bulletin (SB)	Collins TTR-2100-34-14	1	09/04/2025
Maintenance Tip (MT)	34-056	3	07/24/2018
Service Information Letter (SIL)	Rockwell Collins TTR-2100/4100 SIL 16-5	Initial	05/06/2016

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
TCAS Computer	822-2911-002	Rockwell Collins Inc.	822-2911-002

Related Categories

- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- CRC - Component Reliability Collaboration : Collaborative process to eliminate component removals in production and the during the first year of new airplane in service
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message

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